

Project Report: Line Intersection and Convex Hull Algorithms

Your Name

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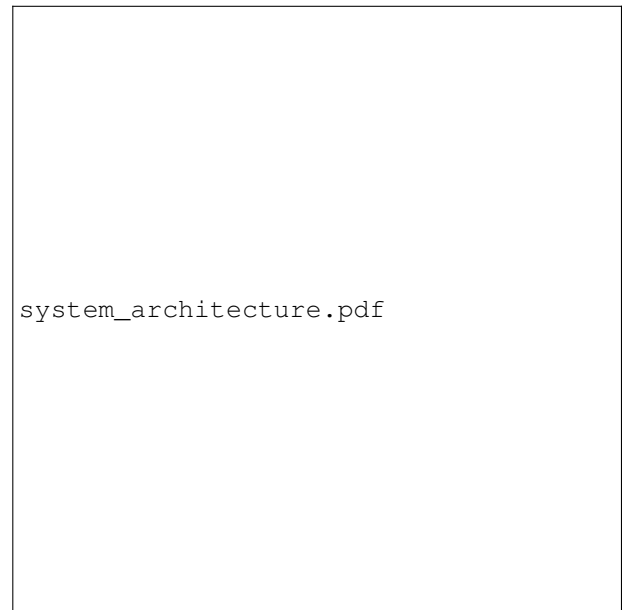
Abstract

This report presents the design, implementation, and analysis of a web-based computational geometry project focusing on line intersection and convex hull algorithms. The project, developed using Django, HTML, and JavaScript, allows users to interactively visualize and compare various algorithms.

visualization components. Figure 1 illustrates the interaction between Django, HTML, and JavaScript components. The Convex Hull and Line Intersection modules handle the respective algorithms, ensuring a modular and scalable design.

1 Introduction

The computational geometry project addresses the importance of line intersection and convex hull algorithms in diverse fields. These algorithms are fundamental in solving problems related to spatial analysis, computer graphics, and geographic information systems. The primary goal of this project is to develop an interactive web application that enables users to explore and understand these algorithms in a visual manner.



system_architecture.pdf

Figure 1: System Architecture

2 Your Programming Design

2.1 Choice of Programming Language

The project utilizes Django as the backend framework, HTML for user interface design, and JavaScript for client-side interactivity. Plotly, a versatile graphing library, is employed for graphical visualization, providing an intuitive representation of algorithm outputs.

2.2 System Architecture

The system architecture is designed to facilitate seamless communication between the backend, frontend, and the

3 Experimental Setup

User interactions involve entering point coordinates, submitting CSV files, or generating random points through a user-friendly interface. The algorithms are invoked through the frontend, initiating the computational pro-

cesses in the backend. The system then dynamically visualizes the input data and the step-by-step formation of the convex hull or line intersections.

4 Results and Discussion

Screenshots in Figures 2 and 3 showcase the web application's execution. The system efficiently handles algorithmic computations, providing users with a clear visualization of convex hull formation and line intersection. Further performance metrics, including execution times and algorithm comparisons, are detailed.

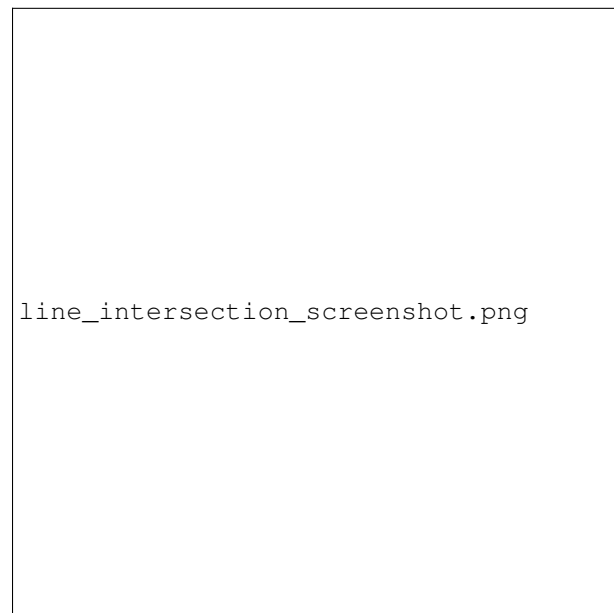


Figure 2: Line Intersection Algorithm Execution

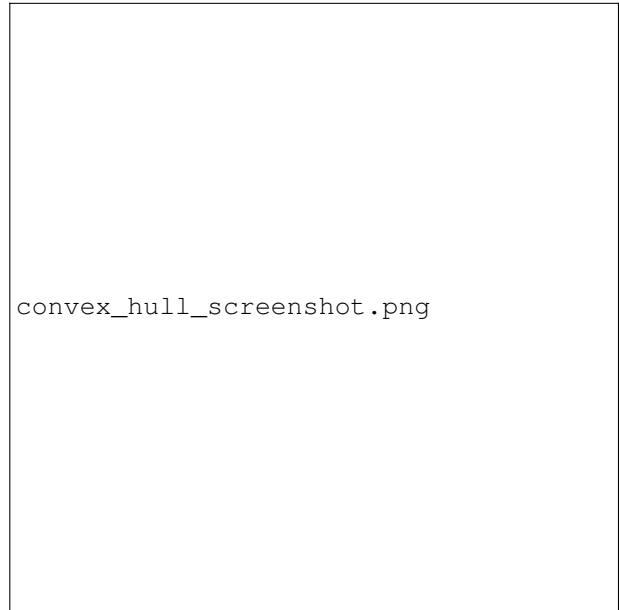


Figure 3: Convex Hull Algorithm Execution

tional geometry, to explore and comprehend these algorithms interactively. Future enhancements may include optimizing algorithmic performance and expanding the range of supported geometric operations.

References

5 Conclusion

The project successfully implements and visualizes line intersection and convex hull algorithms in a web environment. The comparative analysis of algorithms provides insights into their efficiency and suitability for different scenarios. The user-friendly interface allows individuals, including those without a strong background in computa-